

The empirical recurrence for OEIS A205733, proved by transfer matrix

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A205733 carries an empirical recurrence of order 44 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is local to every 2×2 subblock, hence two consecutive lines, so the admissible configurations of such a window are the vertices of a finite digraph and the arrays counted are exactly the walks in it; hence $a(n)$ is a walk count on $S = 43155$ vertices and is C -finite. Whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A205733 is “Number of $(n+1) \times 6$ arrays with every 2×2 subblock having the same number of clockwise edge increases as its horizontal neighbors and no 2×2 subblock having the same number of counterclockwise edge increases as its vertical neighbors.”. It has offset 1 and begins

43155, 80916, 450339, 1302228, 10821462, 33070572, 272838012, 848995332, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 46*a(n-2) - 632*a(n-4) + 2550*a(n-6) + 21009*a(n-8) - 338191*a(n-10) + 2275083*a(n-12) - 9977849*a(n-14) + 32125123*a(n-16) - 80295305*a(n-18) + 160702817*a(n-20) - 261977927*a(n-22) + 350645053*a(n-24) - 385924893*a(n-26) + 348163021*a(n-28) - 255626391*a(n-30) + 151046918*a(n-32) - 70664605*a(n-34) + 25553415*a(n-36) - 6884895*a(n-38) + 1300906*a(n-40) - 153820*a(n-42) + 8568*a(n-44)$ for $n > 51$.

As of the “Last modified” line on the live entry (Oct 09 2015 22:36:28, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The condition the entry imposes is local: it constrains every 2×2 subblock, hence two consecutive lines and nothing beyond. Take as vertices the admissible configurations of one such window and put an edge from u to v whenever v may follow u , which the condition decides by inspection of the two. An admissible array is then exactly a walk, so with M the adjacency matrix on $S = 43155$ vertices and ι, τ the vectors recording which windows may begin and end an array,

$$a(n) = \iota^\top M^{n-1} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C -finite.

3 The proof

Lemma 1. *Let $q(t) = t^{44} - \sum_i c_i t^{44-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+44} - \sum_i c_i M^{j+44-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 46a(n-2) - 632a(n-4) + 2550a(n-6) + 21009a(n-8) - 338191a(n-10) + 2275083a(n-12) - 9977849a(n-14) + 57111111a(n-16) - 271111111a(n-18) + 1111111111a(n-20) - 4444444444a(n-22) + 16666666666a(n-24) - 55555555555a(n-26) + 181818181818a(n-28) - 545454545454a(n-30) + 1515151515151a(n-32) - 4040404040404a(n-34) + 9090909090909a(n-36) - 181818181818181a(n-38) + 363636363636363a(n-40) - 727272727272727a(n-42) + 1454545454545454a(n-44) - 2909090909090909a(n-46) + 5818181818181818a(n-48) - 11636363636363636a(n-50) + 23272727272727272a(n-52) - 46545454545454545a(n-54) + 93090909090909090a(n-56) - 186181818181818181a(n-58) + 372363636363636363a(n-60) - 744727272727272727a(n-62) + 1489454545454545454a(n-64) - 2978909090909090909a(n-66) + 5957818181818181818a(n-68) - 11915636363636363636a(n-70) + 23831272727272727272a(n-72) - 47662545454545454545a(n-74) + 95325090909090909090a(n-76) - 190650181818181818181a(n-78) + 381300363636363636363a(n-80) - 762600727272727272727a(n-82) + 1525201454545454545454a(n-84) - 3050402909090909090909a(n-86) + 6100805818181818181818a(n-88) - 12201611636363636363636a(n-90) + 24403223272727272727272a(n-92) - 48806446545454545454545a(n-94) + 97612893090909090909090a(n-96) - 195225786181818181818181a(n-98) + 390451572363636363636363a(n-100) - 780903144727272727272727a(n-102) + 1561806289454545454545454a(n-104) - 3123612578909090909090909a(n-106) + 6247225157818181818181818a(n-108) - 12494450315636363636363636a(n-110) + 24988900631272727272727272a(n-112) - 49977801262545454545454545a(n-114) + 99955602525090909090909090a(n-116) - 199911205050181818181818181a(n-118) + 399822410100363636363636363a(n-120) - 799644820200727272727272727a(n-122) + 1599289640401454545454545454a(n-124) - 3198579280802909090909090909a(n-126) + 6397158561605818181818181818a(n-128) - 12794317123211636363636363636a(n-130) + 25588634246423272727272727272a(n-132) - 51177268492846545454545454545a(n-134) + 102354536985693090909090909090a(n-136) - 204709073971386181818181818181a(n-138) + 409418147942772363636363636363a(n-140) - 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